

# Design and Analysis of Algorithm (DAA), Lab-01

BTech (CS-B and CE), 3<sup>rd</sup> Semester  
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## Questions

1. **Put them in Order:** Using implementation in C, place the given functions in a list by *increasing order of growth* for sufficiently large values of  $n$ .

|               |              |               |                |
|---------------|--------------|---------------|----------------|
| $n \log_2 n$  | $12\sqrt{n}$ | $\frac{1}{n}$ | $n^{\log_2 n}$ |
| $100n^2 + 6n$ | $n^{0.51}$   | $n^2 - 324$   | $50n^{0.5}$    |
| $2n^3$        | $3^n$        | $2^{32}n$     | $\log_2 n$     |

2. **Fair vs Biased coin:** Using simulation in C, show that the probability of getting a *HEAD* by tossing a *fair coin* is about 0.5. Extend your simulation to compare fair vs biased coin-tossing experiments.

[Interesting Read] Check this article on arXiv titled “*Random Coin Tossing with unknown bias*” for more insight.

3. **Performance analysis of bubble sort:** Using C, implement two different versions of bubble sort simulation for randomised data sequences as follows:

- (i) Bubble sort that terminates if the array is sorted before the  $(n - 1)^{th}$  pass.
- (ii) Bubble sort that always completes the  $(n - 1)^{th}$  pass.

Plot the number of comparisons in both cases to analyse their efficiency.

4. **Towers of Hanoi (ToH):** Simulate the solution to the ToH problem using C. Plot the total number of moves required for solving the problem of  $n$ -discs. What can you conclude about your algorithm from the plot obtained?
  5. **Find the partition point:** Consider an array  $A$  with  $n$  elements containing a run of 0's followed by a run of 1's. Implement a method to find out the exact point of transition between them.
  6. **Element uniqueness:** For given  $n$  random numbers, implement a method in C to check if there are any duplicates. What can you conclude about your method for a sufficiently large value of  $n$ ?
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